

Age and Gender Prediction

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Project: Deep Learning & Generative AI NPPE-1

1. Introduction

This project focuses on building a robust deep learning pipeline for **age and gender prediction** from facial images. Predicting both simultaneously is tricky because facial features change continuously with age, but most models treat age as a categorical label. That doesn't make sense — predicting 25 when the person is 26 shouldn't be considered as wrong as predicting 45.

So, I started with a **simple CNN baseline** and then moved to a stronger **ConvNeXtV2-based model** with several upgrades: **GeM pooling**, **Coral ordinal regression**, **MixUp augmentation**, **label smoothing**, and **gradient checkpointing**. Each change was added for a reason, based on experiments and research from Kaggle competitions and published papers.

In the end, the optimized model clearly outperformed the baseline:

Model	Validation	Public LB
Baseline CNN	0.785	0.811
ConvNeXtV2 + GeM + CORAL	0.8549	0.853

The improvement came not from tuning alone, but from better representation learning and smarter regularization.

2. Dataset and Setup

The dataset had labeled face images with `age` and `gender`. I used **Stratified 5-Fold Cross-Validation**, stratifying by combined `gender + age_group` to keep the folds balanced. Each fold's validation set covered all demographics fairly.

All images were resized to **224×224**, and augmentations were applied using **Albumentations**. Training was done with:

- **Batch Size:** 64
- **Learning Rate:** 1.8e-4
- **Optimizer:** AdamW (weight decay 0.015)
- **Epochs:** 40
- **Precision:** 16-bit mixed precision
- **Gradient Accumulation:** 2 batches
- **Warmup Epochs:** 3
- **Patience:** 4 (early stopping)

I logged everything using **TrackIO** and trained on **2 GPUs (DDP strategy)**.

3. Baseline CNN — What Didn't Work

The baseline CNN had 8 convolutional blocks (32→512 filters), each followed by BatchNorm, ReLU, and MaxPooling. It learned basic texture cues but plateaued early. The main problems were:

- **Limited receptive field** — couldn't capture broader spatial relations.
 - **No pretraining** — had to learn facial features from scratch.
 - **Hard classification for continuous ages** — treated 24 vs. 25 as completely different classes.
- It worked fine for gender (macro-F1 ~0.78) but age predictions were unstable. That's why I needed a pretrained model that already understood general visual features.

4. ConvNeXtV2 Backbone — Why It Helped

I switched to **ConvNeXtV2** (`convnextv2_base.fcmae_ft_in22k_in1k` from TIMM).

This model was trained on **ImageNet-22k**, and is known to balance the structure of CNNs with the performance of Transformers. Facebook's paper ("ConvNeXtV2: Co-Designing and Scaling ConvNets with Masked Autoencoders", 2023) shows that it achieves state-of-the-art transfer results with smaller compute.

For our dataset, ConvNeXtV2 worked much better because:

- It captured fine local features (like wrinkles, skin tone) and global patterns (like face shape).
- Pretraining gave it a strong base for faces, even though the dataset wasn't huge.

This switch alone improved validation RMSE by around 0.04 on average.

5. GeM Pooling — Why Not Just Average Pooling

Instead of plain global average pooling, I used **GeM (Generalized Mean) pooling**, which was first introduced in image retrieval by Radenović et al. (CVPR 2018).

The formula is simple:

$$\text{GeM}(x) = \left(\frac{1}{N} \sum_i x_i^p \right)^{1/p}$$

where p is a learnable parameter.

When $p = 1$, it behaves like average pooling, and as $p \rightarrow \infty$, it becomes max pooling.

So GeM lets the model decide how much to focus on strong local activations vs. overall averages.

In practice, this made the model more sensitive to fine aging cues like wrinkles or jawline changes — which are usually small parts of the image that global average pooling tends to blur out.

6. CORAL Ordinal Regression — Handling Age Properly

For the age head, I replaced the standard linear classifier with **CORAL (Cumulative Ordinal Regression)**, introduced by Cao et al. (Pattern Recognition Letters 2020).

Instead of predicting a single value, it predicts a sequence of binary thresholds: *Is age > 1?*, *Is age > 2?*, ... up to 99 (for 100 age classes).

This structure maintains order among ages and avoids the “jumping class” issue that normal softmax classifiers suffer from.

CORAL has already been used successfully in **Facial Age Estimation** benchmarks like MORPH and UTKFace. Using it here reduced the mean age RMSE from roughly 7.6 → 6.1 years in validation.

7. Data Augmentations

I used a well-researched augmentation strategy from top Kaggle image pipelines:

- **RandomResizedCrop (0.6–1.0)** → handles scale and zoom variation
- **Affine (±25° rotation, ±8% translation)** → pose diversity
- **Color jittering (Brightness/Contrast ±0.35, Hue ±25)** → lighting adaptation
- **Blur/Noise** → robustness to motion or camera blur
- **CoarseDropout (5–12% patch mask)** → simulates occlusions (hair, glasses)

These transforms made the model handle real-world diversity better. Validation stability improved, and overfitting dropped sharply after epoch 10.

8. MixUp + Label Smoothing

To improve robustness, I applied **MixUp (α = 0.25)** and **Label Smoothing (ε = 0.12)**.

- **MixUp** randomly mixes two samples and their labels:

$$\tilde{x} = \lambda x_i + (1 - \lambda)x_j$$

This encourages the model to behave linearly between examples and prevents memorizing specific samples. It was famously used in high-ranking Kaggle competitions like Shopee Product Matching and Cassava Leaf Disease Detection.

- **Label Smoothing** reduces over-confidence and improves calibration, especially for gender classification.

Together, these two techniques improved the final leaderboard score by roughly +0.015 compared to training without them.

9. Gradient Checkpointing — Fixing the Memory Issue

ConvNeXtV2 is heavy, and a batch size of 64 was crashing due to CUDA OOM errors on my GPU. Reducing batch size hurt batch normalization and MixUp effectiveness.

To fix that, I used **gradient checkpointing** — a trick that saves GPU memory by re-computing activations during backprop.

It added ~10-12% more computation per epoch but made training stable and memory-efficient. This method is widely used in large-scale training setups (for example, GPT, ViT, and even in competition notebooks when hardware is limited).

10. Results and Observations

- Validation RMSE dropped significantly and became stable across all folds.
- Gender macro-F1 reached ~0.89 (up from 0.78 baseline).
- The final **harmonic mean score** between normalized age RMSE and gender F1 was **0.8549 (single fold inference)**.
- On the leaderboard, the optimized model reached **0.853**, compared to 0.811 for the baseline CNN.

The improvements were consistent across folds and required fewer epochs to converge due to better pretrained features and regularization.

11. Ethical and Responsible Use

All AI assistance (GPT-5 and Sonnet 4.5) was used only for code structuring, debugging, and research summaries. All design choices, modeling, validation and interpretation were done manually.

12. Conclusion

Switching from a handcrafted CNN to a pretrained ConvNeXtV2 backbone, combined with GeM pooling and CORAL ordinal regression, made the model more interpretable and much more accurate.

Regularization techniques like MixUp, label smoothing, and augmentations helped it generalize across folds.

Gradient checkpointing solved the GPU limitation without downsizing the model.

Overall, the model went from a basic CNN scoring **0.78 → 0.85 (Validation)** and **0.81 → 0.85 (Leaderboard)** — a clear 7-8 % absolute improvement.

This shows how small architectural and training tweaks, backed by solid research, can make a big difference in real performance.

References

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